Reading structured data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

One connected structured-record evidence sheet.

Explain fields, trace contributions, diagnose missing data and report a bounded statistic.

Keep one habit: Keep dataset identity, required/optional policy, observed sum, mean denominator and coverage together.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Dictionaries and records. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Functions and arguments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

Dictionaries and records

> **My remembered idea, example and correction**

Functions and arguments

> **My remembered idea, example and correction**

Branching and boundary tests

> **My remembered idea, example and correction**

W1 · Explain the procedure

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Main cupboard rows (8,3),(6,6) contribute 5 and 0, total 5; required capacity/used fields assumed valid. W2 fixture (4,2),(7,3),(2,1) contributes 2,4,1, total 7. Picture count rows 2/0/missing include 2 measured rows, total 2, mean 1.0, coverage 2/3; add 4→total 6/mean 2.0/coverage 3/4. append retains a real 0. All-empty observed rows give sum 0, denominator 0, mean None; no missing measurement is asserted zero.

  These programs do not parse/validate arbitrary files.

- Reference for this exercise: open this week's student lesson, "Reading structured data", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- In the worked **dataset**, state the number of records and the spare spaces in the second record.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Predict, trace and observe

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Main cupboard rows (8,3),(6,6) contribute 5 and 0, total 5; required capacity/used fields assumed valid. W2 fixture (4,2),(7,3),(2,1) contributes 2,4,1, total 7. Picture count rows 2/0/missing include 2 measured rows, total 2, mean 1.0, coverage 2/3; add 4→total 6/mean 2.0/coverage 3/4. append retains a real 0. All-empty observed rows give sum 0, denominator 0, mean None; no missing measurement is asserted zero.

  These programs do not parse/validate arbitrary files.

- For spare contributions 2, 4 and 1, trace total starting at 0.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Find the first error

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Main cupboard rows (8,3),(6,6) contribute 5 and 0, total 5; required capacity/used fields assumed valid. W2 fixture (4,2),(7,3),(2,1) contributes 2,4,1, total 7. Picture count rows 2/0/missing include 2 measured rows, total 2, mean 1.0, coverage 2/3; add 4→total 6/mean 2.0/coverage 3/4. append retains a real 0. All-empty observed rows give sum 0, denominator 0, mean None; no missing measurement is asserted zero.

  These programs do not parse/validate arbitrary files.

- A loop resets total to 0 on each pass over contributions 2, 4, 1. What final total appears and how should it be fixed?

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · State an honest limit

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Main cupboard rows (8,3),(6,6) contribute 5 and 0, total 5; required capacity/used fields assumed valid. W2 fixture (4,2),(7,3),(2,1) contributes 2,4,1, total 7. Picture count rows 2/0/missing include 2 measured rows, total 2, mean 1.0, coverage 2/3; add 4→total 6/mean 2.0/coverage 3/4. append retains a real 0. All-empty observed rows give sum 0, denominator 0, mean None; no missing measurement is asserted zero.

  These programs do not parse/validate arbitrary files.

- An empty **dataset** produces total 0. Can you conclude that every cupboard is full?

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected independent task

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Trace the named **field**s and missing-value policy. Predict the **output**, state the observed total and mean denominator, and report coverage. What changes if all three rows are empty records?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Preserve records and report measured coverage

```
rows = [{"score": 2}, {"score": 6}, {}]
values = []
for row in rows:
    value = row.get("score")
    if value is not None:
        values.append(value)
observed_total = sum(values)
if len(values) > 0:
    mean = observed_total / len(values)
else:
    mean = None
print(observed_total, len(values), len(rows), mean)
```

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep code and evidence

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Add a labelled reading structured data **evidence** sheet to Python readiness portfolio. Show how you can **process** a list of records consistently.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use W2: Trace all updates for Reading structured data. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use the worked case for Reading structured data. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Additional practice · Earlier W4 transfer

REFERENCE · SECTION 8

Additional practice · Earlier W4 transfer

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Write a loop for records (capacity 9, used 2), (capacity 5, used 1), and (capacity 4, used 4). Predict the total. Reverse the order and test an empty list.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

- Python 3 records: [{"count":2},{"count":0},{}]. Policy: include existing numeric counts, including zero; skip an absent **field**.

- Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

  Use actual Python 3 results to check the **prediction** after reasoning first.

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Included values and absent field

	Count value	Include in mean?
Row 1	2	Yes
Row 2	0	Yes
Row 3	Absent	No

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

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Reading structured data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect three rows

Two rows contain count; the third does not.



2. Accumulate values

The included values sum to 2+0=2, with denominator 2.



3. Report coverage and mean

Mean is 1; measured coverage is 2 of 3 rows.

CHANGE ONE THING

Add {"count":4} as a fourth row.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project portfolio milestone

REFERENCE · SECTION 13

Reference · Project portfolio milestone

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Project: Python readiness portfolio
- Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.
- For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 14

Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

First question and hint

- Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.
- Which labelled step matches this detail? Mean is 1; measured coverage is 2 of 3 rows.
- 1. Inspect three rows
- 2. Report coverage and mean
- 3. Accumulate values
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report coverage and mean Mean is 1; measured coverage is 2 of 3 rows. Structured data need a row policy before aggregation.

Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

Changed-case question and hint

- Changed-case practice: Add {"count":4} as a fourth row. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4. Only the added recorded value contributes; the existing absent **field** remains skipped.

Repair and keep the record

Review Dictionaries and records (python-bridge/10). Goal: Read and update named dictionary **field**s. Short practice: Predict all three original **output**s.

Explain the difference between count=0 and missing price, then predict the lookup after price=0 is added.. Return to this week and explain what you changed.

Review Functions and arguments (python-bridge/8). Goal: Distinguish a function definition from a function call. Short practice: Match every argument to a parameter for the two calls.

Why do **output**s differ, and what prints when both calls are removed?. Return to this week and explain what you changed.

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches.

Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

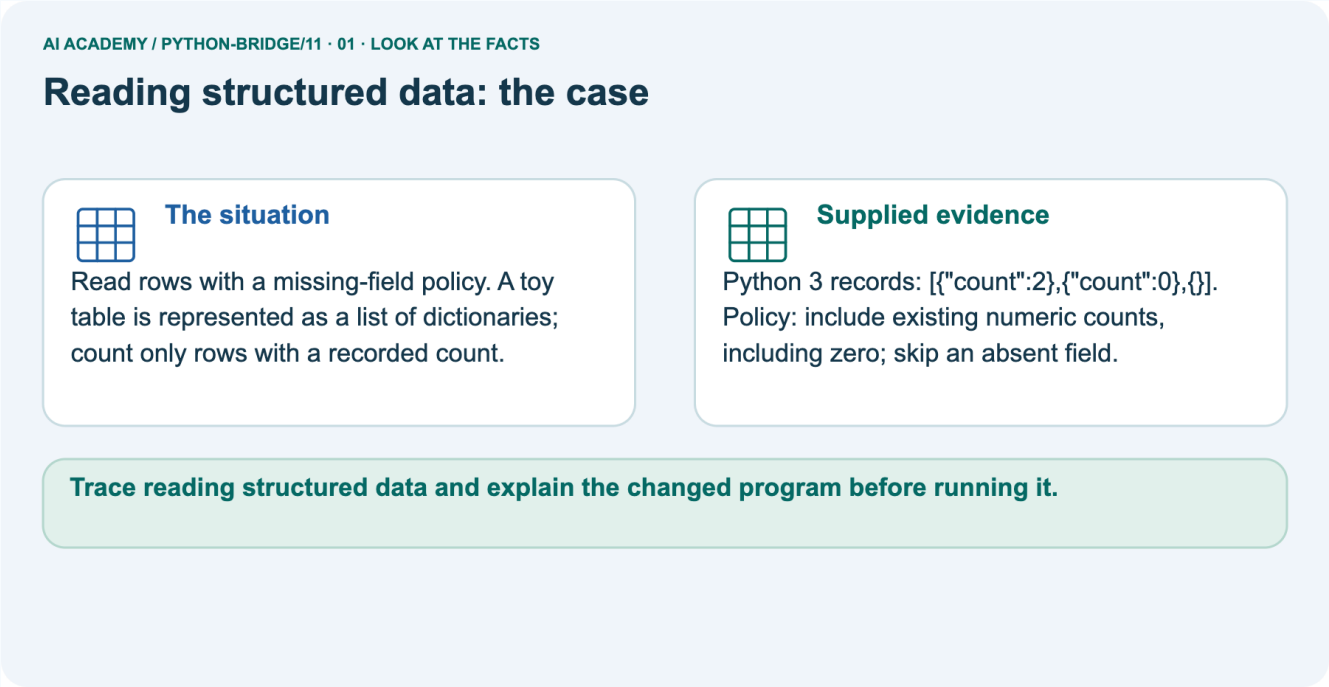
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Use the picture relevant to a current gap.

01. Reading structured data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

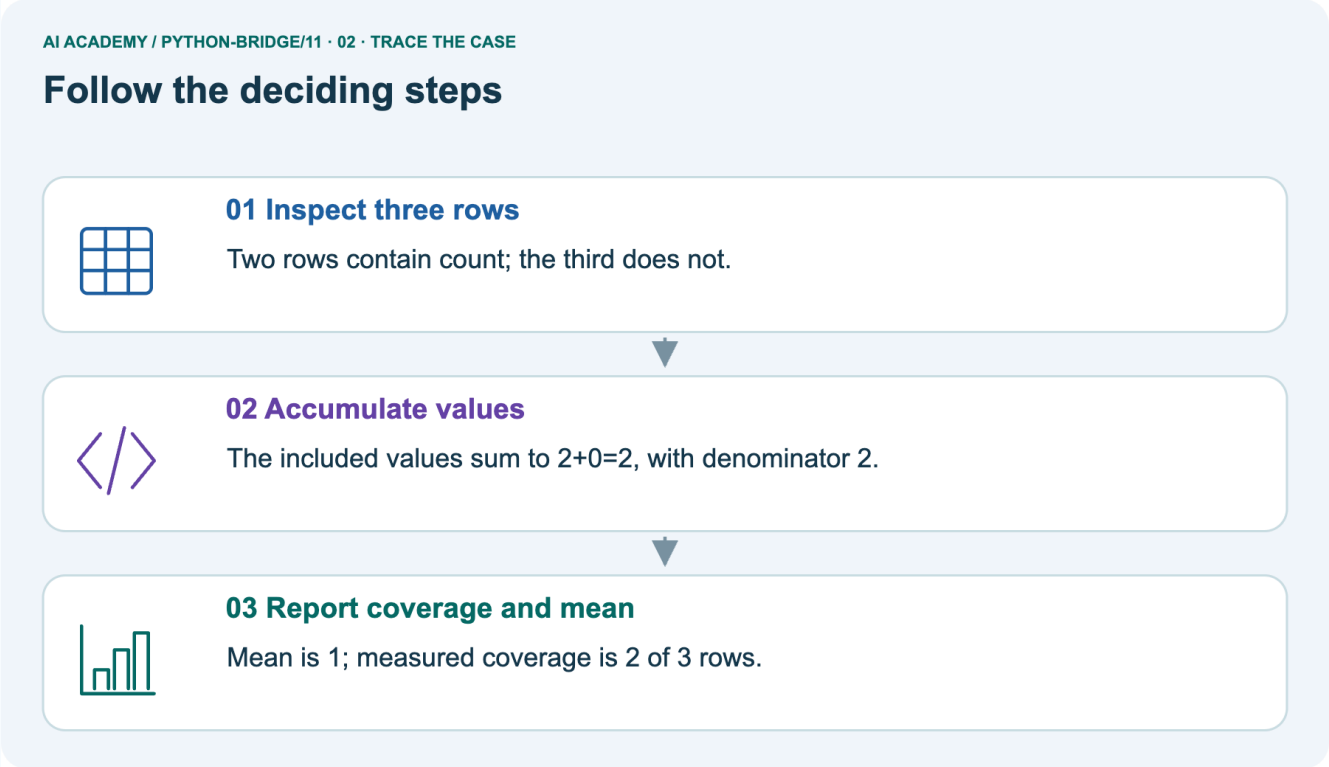
Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count. Supplied facts: Python 3 records: [{"count":2}, {"count":0},{}].

Policy: include existing numeric counts, including zero; skip an absent **field**.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



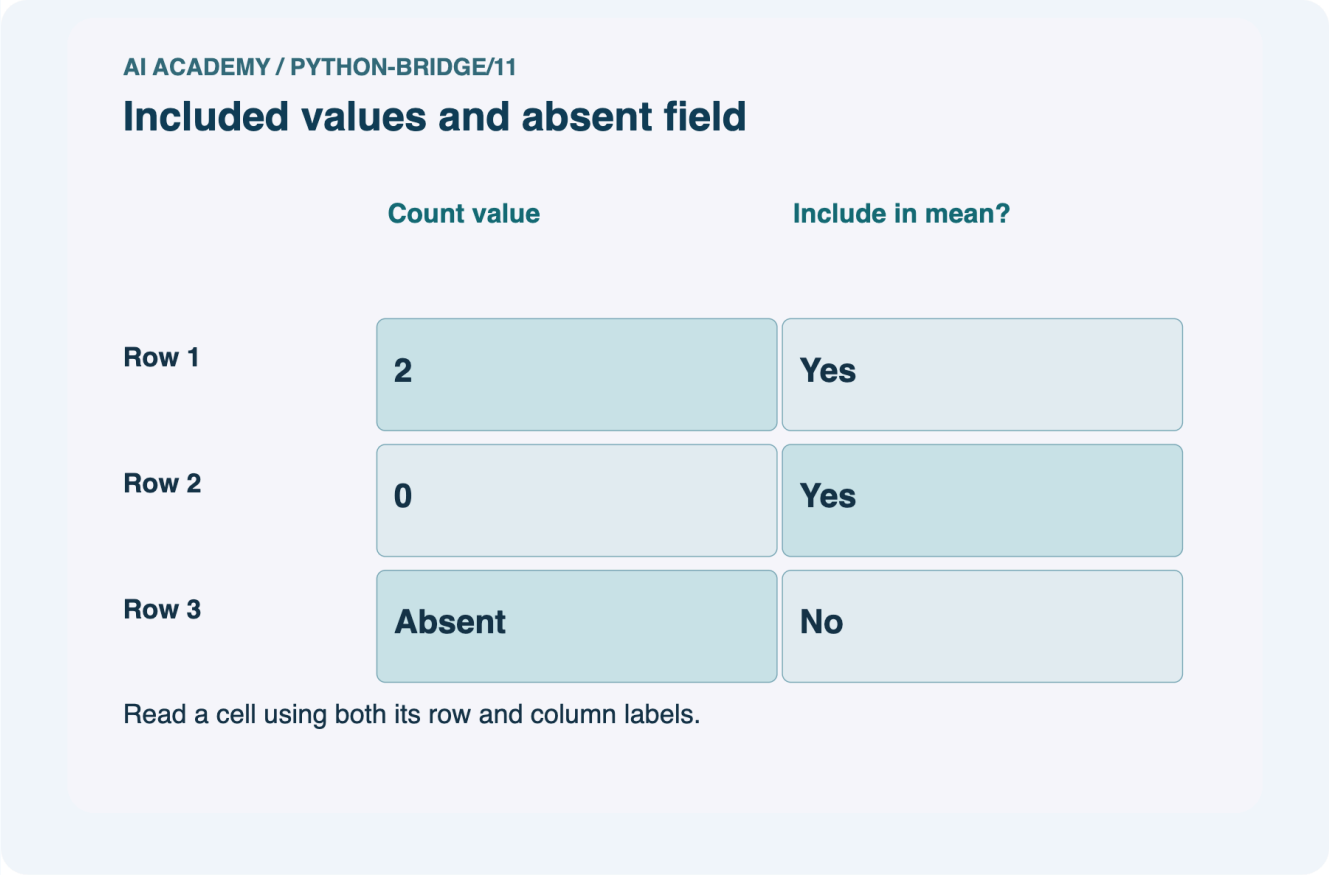
Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect three rows: Two rows contain count; the third does not. → Accumulate values: The included values sum to 2+0=2, with denominator 2. → Report coverage and mean: Mean is 1; measured coverage is 2 of 3 rows.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Included values and absent field



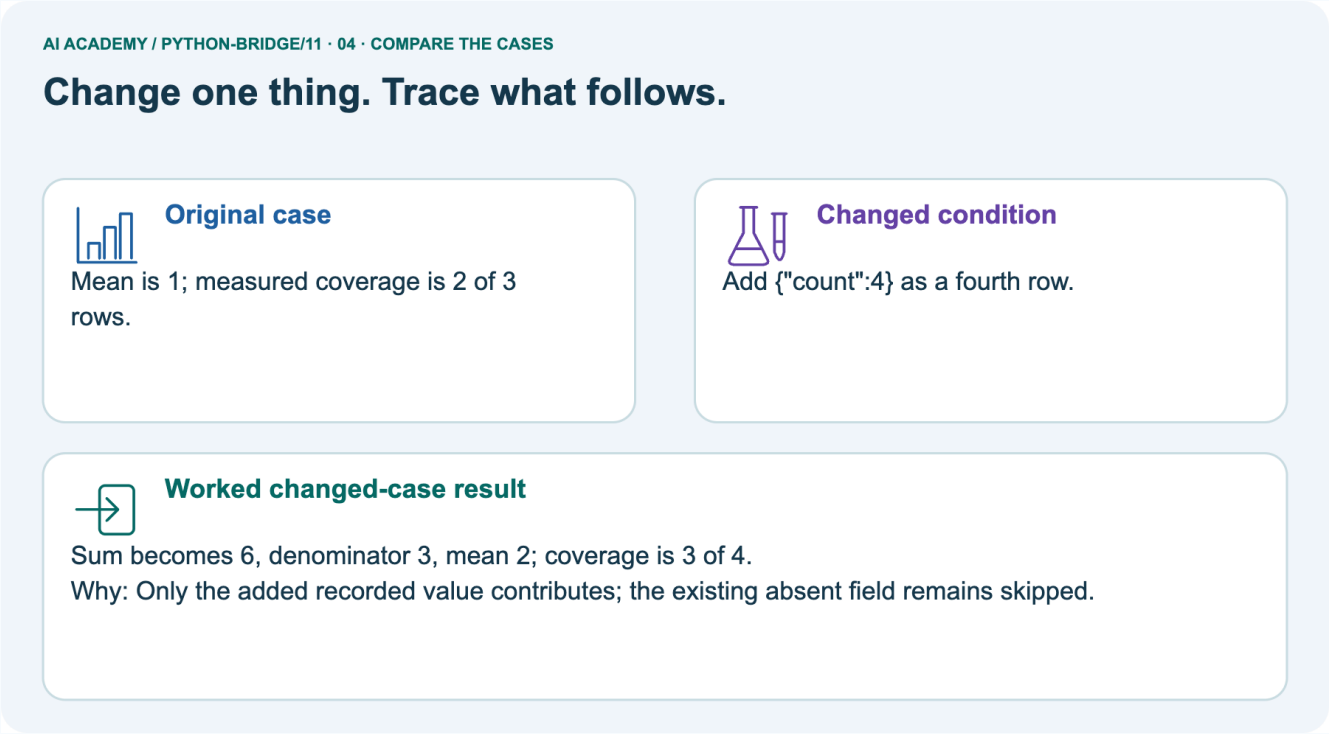
Follow the labelled picture with the matching explanation. Open the vector diagram

Included values and absent **field**. Row 1: Count value = 2, Include in mean? = Yes. Row 2: Count value = 0, Include in mean? = Yes. Row 3: Count value = Absent, Include in mean? = No.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

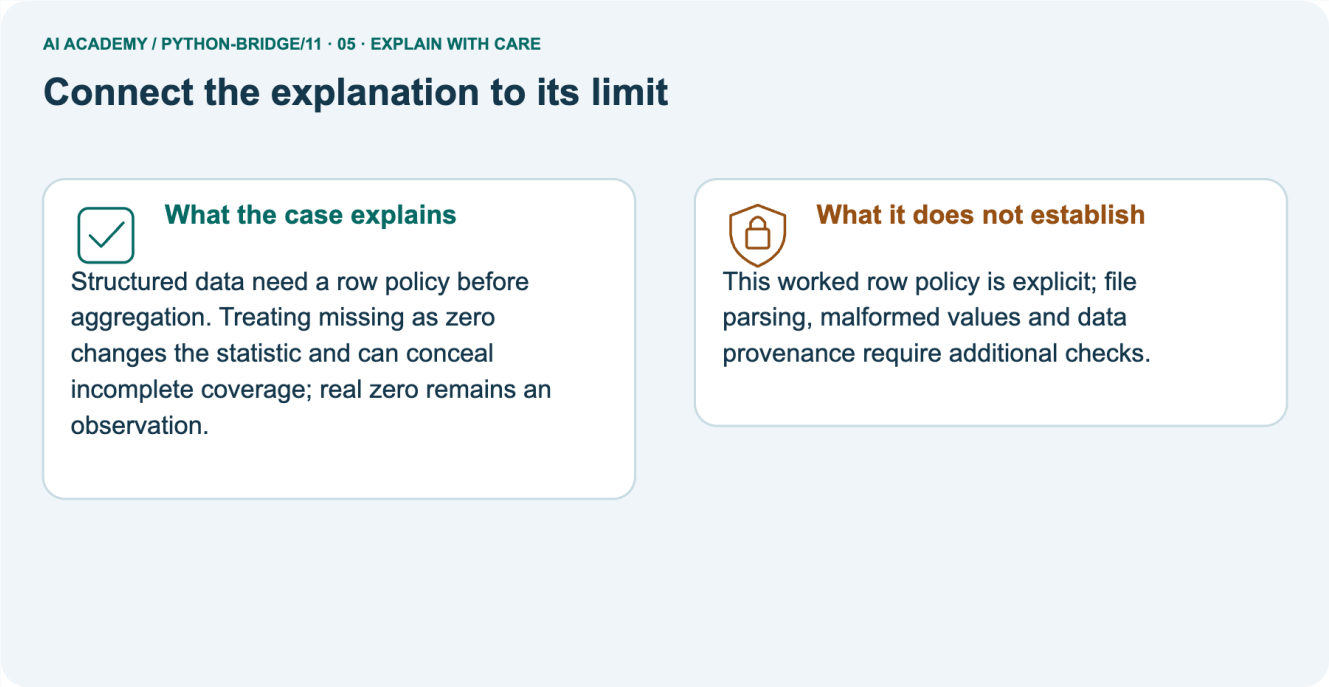
Original case: Mean is 1; measured coverage is 2 of 3 rows. Changed condition: Add {"count":4} as a fourth row. Expected result: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

Reason: Only the added recorded value contributes; the existing absent **field** remains skipped.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

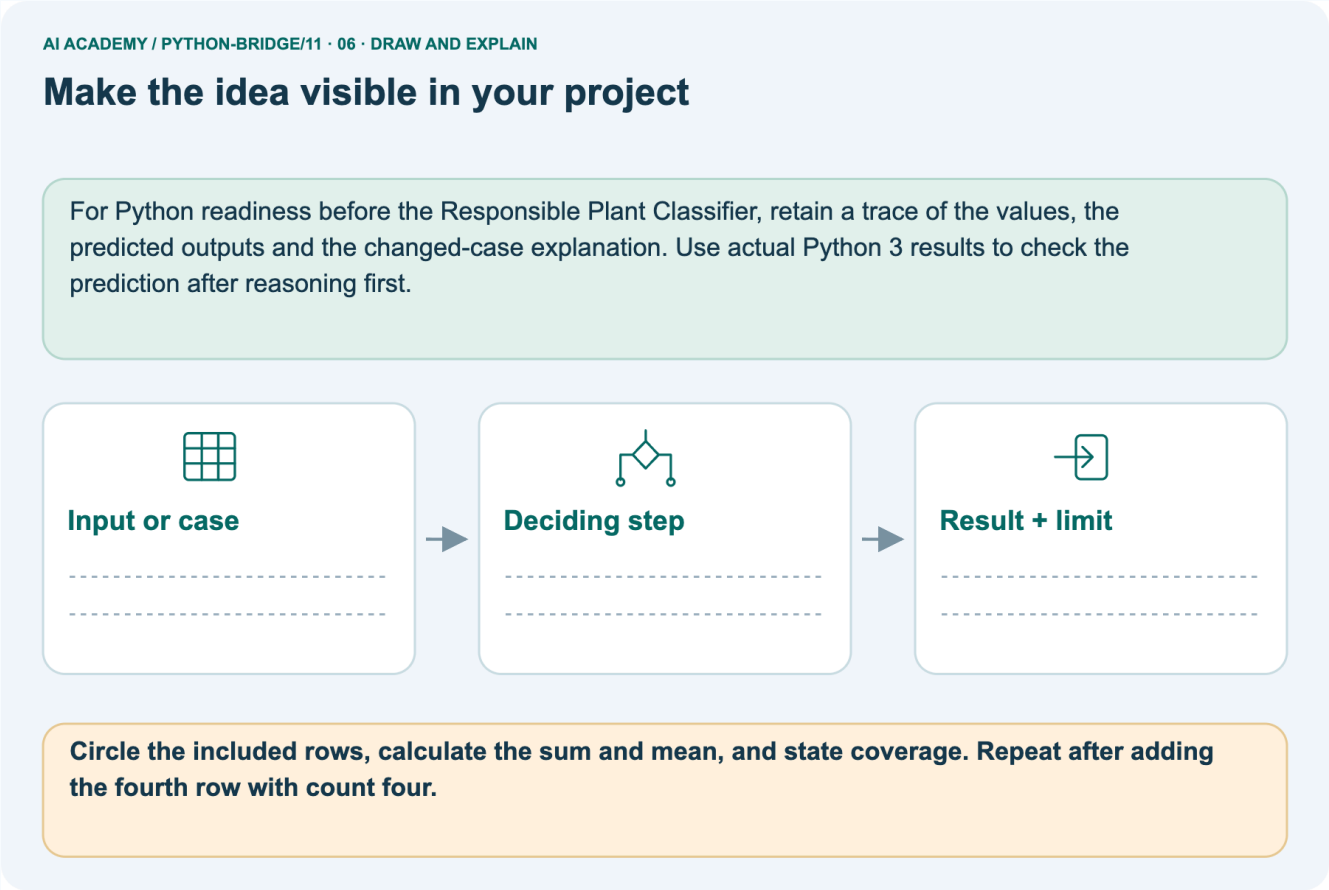
Structured data need a row policy before aggregation. Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

Limit: This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Reading structured data: the case

02. Follow the deciding steps

03. Included values and absent field

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
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<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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